

A/B Testing & Hypothesis Testing Report

Website Checkout Experiment — Control A vs Variant B

Dataset: checkout_ab_test_v2.csv | 1,200 visitor-level observations | Synthetic experiment data

1. Executive Summary

Variant B produced a conversion rate of **11.83%** compared with **9.50%** for Control A. The observed absolute lift was **2.33%**, corresponding to a relative lift of **24.6%**. However, the two-sided two-proportion Z-test returned **p = 0.1905**, so the observed difference is not statistically significant at the 5% level. The 95% confidence interval for the conversion-rate difference (B – A) is **[-1.16%, 5.82%]**, which includes zero.

Executive decision memo: Under a conventional $\alpha = 0.05$ decision rule, retain Control A rather than treating Variant B as a validated improvement. B shows a promising observed lift, but this experiment does not provide sufficient statistical evidence to conclude that the lift is real rather than sampling variation.

2. Hypotheses

H₀: The true conversion rates for A and B are equal ($p_B - p_A = 0$).

H_a: The true conversion rates differ ($p_B - p_A \neq 0$).

Significance level: $\alpha = 0.05$. Test: two-sided two-proportion Z-test.

3. Sample Size & Conversion Rates

Metric	Control A	Variant B
Visitors	600.0	600.0
Conversions	57.0	71.0
Conversion rate	9.50%	11.83%
95% CI for rate	[7.15%, 11.85%]	[9.25%, 14.42%]

The achieved sample size is 600 visitors per arm. A pre-experiment required sample size cannot be determined from the supplied task alone because no baseline conversion rate, target minimum detectable effect, power, or allocation assumptions were specified.

4. Statistical Test Results

Statistic	Result
Absolute lift (B – A)	2.33%
Relative lift	24.6%
Pooled conversion rate	10.67%
Z statistic	1.3092
P-value	0.1905
95% CI for difference	[-1.16%, 5.82%]
Chi-square statistic	1.7141
Chi-square p-value	0.1905

Interpretation: because $p > 0.05$ and the confidence interval crosses zero, the experiment does not reject **H₀** at the 5% significance level.

5. Secondary Checkout Funnel View

Variant	Checkout starts	Conversions	Checkout→conversion rate
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A	186	57	30.65%
B	194	71	36.60%

6. Business Interpretation & Limitations

The observed B lift is directionally positive, but statistical uncertainty is substantial enough that the data does not establish a reliable conversion improvement. Results should be interpreted only for the experiment population represented by this synthetic dataset. Before production rollout, validate random assignment, experiment duration, sample-ratio consistency, tracking quality, and guardrail metrics such as revenue per visitor and checkout errors.

7. Reproducibility

The accompanying Jupyter Notebook reproduces data validation, conversion-rate calculations, the two-proportion Z-test, confidence interval, chi-square test, and the executive decision logic.